

Policy Cross-reference of artifacts and reviews that need to be created and reviewed in each lifecycle phase (Discover, Define, Design, Develop, Delivery, and Run) for projects for an aircraft manufacturer

The matrix cross-references the required governance artifacts and review gates across each phase of the aircraft development lifecycle—from early market research to in-service support.

Lifecycle Phase	Required Artifacts	Mandatory Review Gates
1. Discover	• Market Analysis & Customer Needs Document • Initial Business Case & Strategy Alignment • Feasibility & Trade Studies • Initial Regulatory Strategy (FAA/EASA baseline)	• Gate 0 (G0): Project Justification • Portfolio Alignment Review

2. Define	• Concept of Operations (ConOps) • Top-Level Aircraft Requirements (TLARs) • Initial Certification Plan (CP) • Master Program Schedule & WBS • Initial Risk Register & Budget Allocation	• Gate 1 (G1): Concept Readiness • System Requirements Review (SRR) • System Functional Review (SFR)
3. Design	• Systems Architecture & Interface Control Documents (ICDs) • Functional Hazard Assessment (FHA - ARP4761A)	• Gate 2 (G2): Design Freeze • Preliminary Design Review (PDR) • Critical Design Review (CDR)

	• Software/Hardware Development Plans (DO-178C/DO-254) • Preliminary & Detailed Engineering Drawings • Compliance Checklists (Parts 23/25)	• Safety & Airworthiness Review
4. Develop	• First Article Inspection (FAI - AS9102) Reports • Ground & Flight Test Plans / Test Procedures • Prototype Airframe & Avionics Assemblies • Manufacturing Master Plan & Tooling Setup	• Gate 3 (G3): Manufacturing Release • Test Readiness Review (TRR) • Flight Test Readiness Review (FTRR) • Change Control Board (CCB) Reviews

	• Change Control Logs (ECRs/ECOs)	
5. Delivery	• Type Certificate (TC) & Production Certificate (PC) • Flammability & Interior Completion Certificates • Airplane Flight Manual (AFM) & Maintenance Manuals • Certificate of Airworthiness (CoA) per Aircraft • Customer Acceptance Sign-off & Delivery Checklist	• Gate 4 (G4): Certification & Entry into Service (EIS) • Final Certification Board Review • Customer Acceptance Review

6. Run	• Continued Airworthiness Monitoring Plan (CAMP) • Service Bulletins (SBs) & Airworthiness Directives (ADs) • Reliability & Fleet Performance Metrics • Customer Support & Spares Allocation Plans • Post-Implementation Audit Report	• Gate 5 (G5): Post-Launch Review • Continued Airworthiness Review Board (CARB) • Annual Portfolio Performance Audit
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Key Phase Transitions & Decision Rules

- **Define $\rightarrow$ Design:** Project cannot proceed to Detailed Design without approved TLARs and a baseline Certification Plan submitted to the regulatory authority (FAA/EASA).
- **Design $\rightarrow$ Develop:** Physical manufacturing of flight-test articles requires CDR approval and signed-off Development Assurance Levels (DALs per ARP4754B).
- **Develop $\rightarrow$ Delivery:** Flight testing must conclude with full Means of Compliance (MoC) verification before Type Certification (TC) issuance.
- **Delivery $\rightarrow$ Run:** Individual aircraft delivery requires a Certificate of Airworthiness (CoA) and complete compliance sign-offs for custom luxury cabin configurations (FAR 25.853).